Programming and Spreadsheet Modeling

BI-5050 Business Decision

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Front Range Outfitters (FRO)
Decision: Retail Location Expansion

- Front Range Outfitters (FRO) is a Colorado-based outdoor recreation retailer selling hiking, camping, cycling, climbing, and seasonal outdoor products. The company currently operates several retail stores along Colorado's Front Range and has grown steadily over the past several years.

- FRO's leadership believes there is an opportunity to open one additional retail location, but the company has limited capital and wants a more rigorous analysis before committing to the expansion.

- Decision: Should FRO open a new retail location and where?

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Front Range Outfitters (FRO)
Retail Location Alternatives

- Management has identified three candidate locations:
  - Golden
  - Westminster
  - Parker
- The locations differ substantially in expected customer traffic, lease costs, labor costs, market characteristics, store size, and startup investment.
- The company's initial question sounds simple: Which location should Front Range Outfitters select for its next store?
- But management does not yet have enough information to make that decision.

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Front Range Outfitters (FRO)
Business Analytics Team

- You are part of the Business Analytics team at Front Range Outfitters.
- The executive team has asked your group to evaluate the proposed expansion and ultimately recommend whether the company should open another store and, if so, which candidate location should be selected.
- Management expects more than a static report. At the end of the project, you will present an interactive Excel decision-support model that executives can use during the decision meeting to examine your recommendation and test alternative assumptions.
- The analysis will develop as additional data and management requirements become available.

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Front Range Outfitters (FRO)
Additional Financial Analysis

- The Business Analytics team has developed break-even and discounted cash flow (DCF) models for the three proposed locations.
- Before proceeding with the expansion decision, Finance wants a comparable financial analysis of FRO's 10 existing retail locations.
- Finance has provided operating and investment data for these locations.
- Management wants to understand the profitability and financial performance of the existing stores to provide additional context for the expansion decision.

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Front Range Outfitters (FRO)
Financial Evaluation of Existing Stores

- Using the financial data provided, develop a Python program to evaluate all 10 existing locations.
- Calculate break-even transactions, annual operating cash flow, and net present value (NPV) for each location.
- Identify locations that are operating above break-even but have a negative NPV.
- Export the analytical results to Excel for review by Finance.

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